input_ids

[B, S]

model · Qwen4ExpTextModel

≈125B

aux_input_ids

input_ids

[B, S]

[B, S]

[B, S]

Embedding

embed_tokens · 248320 → 2048 · 509M

[B, S, 2048]

ple_input_ids

hidden_states

layers · Qwen4ExpTextDecoderLayer

× 48

≈124B

[B, S]

Qwen4ExpTextPLELayer

ple · ≈32.8M

+

Qwen4ExpTextGatedResidual

attn_hyper_connection · ≈6.30M

Qwen4ExpTextGatedDeltaNet

linear_attn · ≈50.0M

Qwen4ExpTextAttention

self_attn · 11.0M

×

+

Qwen4ExpTextGatedResidual

mlp_hyper_connection · ≈6.30M

Qwen4ExpTextSparseMoeBlock

mlp · 2.52B

×

+

hidden_states

Qwen4ExpTextGatedResidual

hyper_connection_final · 6.30M

last_hidden_state

Linear

lm_head · hidden_size → vocab_size

logits

Load Balancing Loss Func

+